26900352 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEMAT303 - EXPLORING THE HARMONY OF AUTOMATA

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **2** marks

1. What is a subgraph? Give one example. [U] / [CO1]
2. Define a sequence. How does a sequence differ from a set?? [U] / [CO1]
3. Write a regular expression describing all strings over $\{0, 1\}$ that end in 1. [A] / [CO2]
4. Define the transition function of a DFA, stating its domain and its range. [K] / [CO2]
5. What is meant by a sentential form? [K] / [CO3]
6. State the relationship between CFGs and PDAs. [K] / [CO3]
7. Can every decidable language be enumerated? Justify. [U] / [CO4]

8. State the significance of a Universal Turing Machine. [U] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Define alphabet, string, length of a string, the empty string, the reverse of a string and a substring, giving examples over $\sum = \{0, 1\}$ and over $\sum = \{a, b, c\}$. [U] / [CO1]

10. Explain the difference between a function and a relation, and show that every function is a special kind of relation. Give the table of $f : \{0, 1, 2, 3, 4\} \to \{0, 1, 2, 3, 4\}$ defined by $f(n) = n + 1$ mod 5 and determine its range. [A] / [CO1]

11. Construct an NFA using epsilon-transitions that recognizes $A = 0^k : \text{ k is a multiple of 2 or a multiple of 3},$ and describe its computation on the input 000000. [A] / [CO2]

12. Describe the procedure CONVERT(G) for converting a GNFA into an equivalent regular expression, explaining the recursion and the base case. [U] / [CO2]

13. Explain instantaneous descriptions of a PDA with an example. [A] / [CO3]

14. Explain the importance of selecting an appropriate string in proving that a language is not context-free using pumping lemma [A] / [CO3]

15. Explain the formal definition of a Turing Machine. [U] / [CO4]

16. Explain the role of enumerators in Turing-recognizable languages. [U] / [CO4]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. a) Define the operations of union, intersection, difference, complement and symmetric difference on sets. [A] / [CO1]
    b) Verify the distributive laws $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, for $A = \{1, 2, 3\}, B = \{2, 3, 4\}$ and $C = \{3, 4, 5\}$.

18. a) Convert the following NFA into a DFA by the subset construction: states $\{1, 2, 3\}$, start state 1, accept state 1, with transitions $\delta(1, b) = 2, \delta(1, \epsilon) = 3, \delta(2, a) = 2, 3, \delta(2, b) = 3$ and $\delta(3, a) = 1$. b) Compute all the necessary epsilon-closures, list the states of the 8-state DFA, draw the reachable part of the machine and describe the language recognized. [A] / [CO2]

19. Prove that none of the following languages is regular: (i) $\{0^n 1^n : n \geq 0\}$; (ii) $\{ww : \text{w is a string over } \{0, 1\}\}$; (iii) $\{1^{n^2} : n \geq 0\}$. In each case exhibit the string chosen and examine every possible decomposition. [An] / [CO2]

20. Let $D = \{xy \mid x, y \in \{0, 1\}^*, \ |x| = |y|, \ x \neq y\}$.Show that D is a context free language. [A] / [CO3]

21. Explain nondeterministic Turing machines and show that they can be simulated by deterministic Turing machines. [An] / [CO4]

22. Explain the equivalence of different models of computation with Turing Machines and how simulation establishes equivalence. [An] / [CO4]

[10x3 = 30]